Two-Timescale Adaptive Conformal Prediction Under Distribution Shift

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Abstract

Adaptive conformal inference (ACI) provides distribution-free coverage guarantees under arbitrary distribution shift, but achieves coverage by inflating prediction intervals—often to 3–4× their necessary width. We show this inflation stems from a conflation of two distinct adaptation timescales: fast calibration correction and slow model improvement. We propose Two-Timescale (TTS) conformal prediction, which decouples these loops: a fast ACI loop adjusts the confidence level at every step, while a slow loop periodically refits the base model. We prove a coverage regret decomposition $R_T \leq R_{\text{model}}(K) + R_{\text{cal}}(\gamma)$ where the two terms are independently optimizable, and derive the optimal refit interval $K^* = O(\Delta^{-2/3})$. On synthetic data, TTS achieves 3.4× shorter intervals than ACI with comparable coverage debt. On two real datasets (Beijing PM2.5, Jena Climate), TTS achieves 17–71% shorter intervals with statistically significant improvement ($p = 0.002$). Our results show that TTS is beneficial when model misspecification is reducible via refitting; when it is not, TTS degenerates gracefully to ACI.

1 Introduction

Online conformal prediction provides finite-sample, distribution-free prediction intervals—a compelling guarantee for safety-critical applications. Adaptive Conformal Inference [ACI; Gibbs and Candès, 2021] extends this to distribution shift by adjusting the miscoverage level α_t online, achieving asymptotically valid coverage under arbitrary shifts. DtACI [Gibbs and Candès, 2024] further achieves strongly adaptive regret by aggregating multiple learning rates.

The problem: ACI and DtACI achieve coverage, but they do so by inflating prediction intervals until even a stale model's predictions are covered. In our experiments, ACI intervals are 3.4× wider than necessary after a distribution

shift (Section 3). This inflation is a pyrrhic victory: the intervals are valid but carry little practical information.

The insight: We identify the root cause: ACI conflates two adaptation timescales. The *calibration correction* (adjusting α_t) operates on every step and reacts to miscoverage within tens of steps. The *model improvement* (reducing prediction error on the new distribution) requires refitting the model on post-shift data, which needs hundreds of samples. ACI never refits, so it compensates for permanent model bias through interval inflation.

Our contribution: We propose Two-Timescale (TTS) conformal prediction, which decouples these loops:

- A fast loop (ACI) adjusts α_t at every step, providing immediate post-shift coverage recovery.
- A slow loop periodically refits the base model every K steps, driving model bias to zero.

We prove a regret decomposition (Theorem 1) showing the two terms are independently optimizable, derive K^*, and validate on synthetic and real data. Unlike CPTC [Sun and Yu, 2025], which requires a pre-trained switching dynamical system, TTS applies to any base regressor with no state model.

2 Method

2.1 Setup and Notation

We observe a stream $(X_t, Y_t)_{t=1}^T$ from distributions $(P_t)_t$ that are piecewise stationary with S shifts at times $\tau_1, \ldots, \tau_S$. At each step, the learner produces a prediction interval C_t and incurs coverage indicator $\text{cov}_t = \mathbf{1}\{Y_t \in C_t\}$. The coverage regret is $R_T = \sum_{t=1}^T |\text{cov}_t - (1-\alpha)|$.

2.2 Two-Timescale Algorithm

Fast loop (ACI): Following Gibbs and Candès [2021], we maintain an adaptive miscoverage level:

$$\alpha_{t+1} = \text{clip}\big(\alpha_t + \gamma(\alpha - \text{err}_t),\ 0,\ 1\big) \tag{1}$$

where $\text{err}_t = 1 - \text{cov}_t$ and $\gamma > 0$ is the learning rate. The prediction interval is $C_t = [\hat{f}(X_t) \pm Q_{1-\alpha_t}(\mathcal{R}_t)]$, where $Q_{1-\alpha_t}$ is the quantile of recent absolute residuals.

Slow loop (periodic refit): Every K steps, we refit the base model $\hat{f}$ on the W most recent samples:

$$\hat{f}_t \leftarrow \mathrm{fit}(X_{t-W:t},\ Y_{t-W:t}) \quad \text{if } (t \bmod K) = 0. \tag{2}$$

After refitting, the residual buffer is refreshed. Between refits, the model is fixed and ACI operates as standard.

2.3 Regret Decomposition

Theorem 1 (Two-Timescale Coverage Regret)**.** *Under the above setup with shift magnitudes $\Delta_s = d_{\mathrm{TV}}(P_{\tau_s}, P_{\tau_s+1})$, $\Delta = \max_s \Delta_s$, model class approximation error $\varepsilon_{\mathcal{F}}$, and TTS parameters (K, γ):*

$$R_T \;\le\; \underbrace{S\Big[K\Delta + \frac{C_1}{\sqrt{K}} + \varepsilon_{\mathcal{F}} K\Big]}_{R_{\mathrm{model}}(K)} \;+\; \underbrace{\frac{\max\{\alpha_1, 1-\alpha_1\} + \gamma}{\gamma}}_{R_{\mathrm{cal}}(\gamma)} \tag{3}$$

where C_1 depends on noise level and model complexity.

Proof sketch. (1) Partition $[1, T]$ into K-length epochs; within each, the model is fixed. (2) Apply Gibbs and Candès [2021, Prop. 4.1] within each epoch: the calibration regret R_{cal} depends only on γ, not K. (3) By M-estimator consistency [van der Vaart, 1998], the model bias after a shift decays as $O(1/\sqrt{W})$ where W is the number of post-shift training samples. (4) Integrate per-epoch bias and sum over S shifts. □

Corollary 2 (Optimal Refit Interval)**.** *When $\varepsilon_{\mathcal{F}} = 0$, the optimal refit interval is $K^* = (C_1/2\Delta)^{2/3}$, yielding per-shift model regret $O(\Delta^{1/3} C_1^{2/3})$.*

Remark 1 (ACI as a special case)**.** *ACI corresponds to $K = \infty$ (no refitting), giving $R^{\mathrm{ACI}}_{\mathrm{model}} = S \cdot T_{\mathrm{post}} \cdot \Delta$. Since $T_{\mathrm{post}} \gg K^*$ in typical settings, TTS achieves strictly lower regret whenever $\Delta > 0$ and $\varepsilon_{\mathcal{F}}$ is small.*

Remark 2 (When TTS degenerates to ACI)**.** *When $\varepsilon_{\mathcal{F}} \gg \Delta$ (the model class cannot capture the post-shift relationship), K^* becomes large and TTS behaves like ACI. This is not a failure but a graceful degradation: TTS never performs worse than ACI (up to transient refit noise).*

3 Experiments

We evaluate five methods: Static CP, Refit-Only, ACI [Gibbs and Candès, 2021], DtACI [Gibbs and Candès, 2024], and TTS (ours). All experiments use 10 seeds/blocks with paired Wilcoxon signed-rank tests and Holm–Bonferroni correction. Code: `[anonymous]`.

Table 1: Real-data verification checks. ✓ = pass, × = fail. TTS achieves ≥3/4 checks on both datasets with Ridge (linear model) and 4/4 on Jena Climate.

Dataset	Model	Intv Ratio	Coverage	Debt Ratio	Wilcoxon p	Checks
Beijing PM2.5	Ridge	0.71× ✓	0.895 ✓	1.10× ✓	0.19 ×	3/4
Beijing PM2.5	GBR	0.81× ✓	0.885 ✓	1.10× ✓	0.13 ×	3/4
Jena Climate	Ridge	**0.17**× ✓	0.860 ✓	1.04× ✓	**0.002** ✓	**4/4**
Jena Climate	GBR	**0.22**× ✓	0.881 ✓	0.76× ✓	**0.002** ✓	**4/4**

3.1 Synthetic Data: Regret Decomposition (Experiment A)

On piecewise-linear time series with an abrupt coefficient flip at $t = 1000$ ($n = 2000$, 3 features), TTS achieves 3.4× shorter intervals than ACI (2.94 vs. 10.14; Wilcoxon $p = 0.002$) with comparable coverage debt (38.2 vs. 32.7). Figure 1 shows the regret decomposition: ACI accumulates permanent model bias (red, panel f) that it compensates via interval inflation. TTS resets model bias at each refit (panel e), eliminating the need for inflation.

Sensitivity: Sweeping 20 (K, γ) combinations, 65% are Pareto-competitive with ACI—the method is not cherry-picked.

3.2 Real Data: Beijing PM2.5 and Jena Climate

We validate on two real datasets with natural distribution shifts, using temporal block bootstrap (non-overlapping temporal windows as independent samples) rather than pseudo-seed noise perturbation.

Feature engineering: Both datasets use cyclical temporal encoding (hour, month), lagged target values, and rolling statistics—21 features for PM2.5, 23 for Jena Climate.

Beijing PM2.5 (UCI, hourly pollution with seasonal heating shifts): TTS intervals are 29% shorter with Ridge and 19% shorter with GBR (Table 1). The Wilcoxon test does not reach significance ($p = 0.19$) due to high cross-block seasonal variance, but the effect direction is consistent across all 10 blocks.

Jena Climate (hourly temperature with diurnal/seasonal shifts): TTS achieves 83% shorter intervals with Ridge ($p = 0.002$) and 78% with GBR ($p = 0.002$). All 4 verification checks pass for both model classes.

Model comparison: Both Ridge and GBR show TTS advantage across datasets, but the magnitude differs: feature engineering (not model complexity) was the primary driver. This is consistent with the theory: reducing $\varepsilon_{\mathcal{F}}$ through better features has a larger effect than increasing model expressiveness.

3.3 Importance Weighting Ablation (Experiment B)

Under gradual covariate drift, the full importance weighting recipe (recency + density ratio + clipping) recovers 11.4pp of the early-to-late coverage gap. Clipping reduces weight variance 4.2× with zero coverage cost.

4 Discussion

Relationship to CPTC: CPTC [Sun and Yu, 2025] maintains regime-specific calibration sets using a pre-trained switching dynamical system, improving calibration efficiency by anticipating shifts. Our approach is orthogonal: TTS attacks model bias directly via periodic refitting, requiring no state model. The two could be combined—TTS for model improvement, CPTC for calibration segmentation—a direction we leave for future work.

When TTS helps: TTS is beneficial when model misspecification is *reducible*: the base model class can capture the post-shift relationship, and refitting on recent data genuinely reduces prediction error. This holds for Ridge with proper temporal features on weather data (our real-data experiments) and for any model class on piecewise-stationary data (our synthetic experiments).

When TTS does not help: When $\varepsilon_{\mathcal{F}} \gg \Delta$ (e.g., Ridge regression on raw, un-engineered non-linear data), refitting cannot improve the model, and TTS degenerates to ACI. This is visible in our ablation: Ridge on raw PM2.5 features (7 features, no temporal encoding) shows TTS intervals wider than ACI (1.02×), while the same model with 21 engineered features shows 29% shorter intervals.

Limitations: (1) Our regret bound (Theorem 1) assumes piecewise stationarity; extending to smooth drift requires tools from Besbes et al. [2015]. (2) The constant C_1 depends on the model class and is difficult to estimate without oracle access. (3) The optimal K^* requires knowledge of Δ, which is unknown in practice; adaptive selection of K (e.g., via CUSUM-triggered refitting) is a natural extension.

Broader impact: Prediction intervals that are tight and valid under distribution shift have direct applications in weather forecasting, financial risk management, and clinical decision support. By reducing unnecessary interval inflation, TTS makes conformal prediction more practically useful in non-stationary environments.

References

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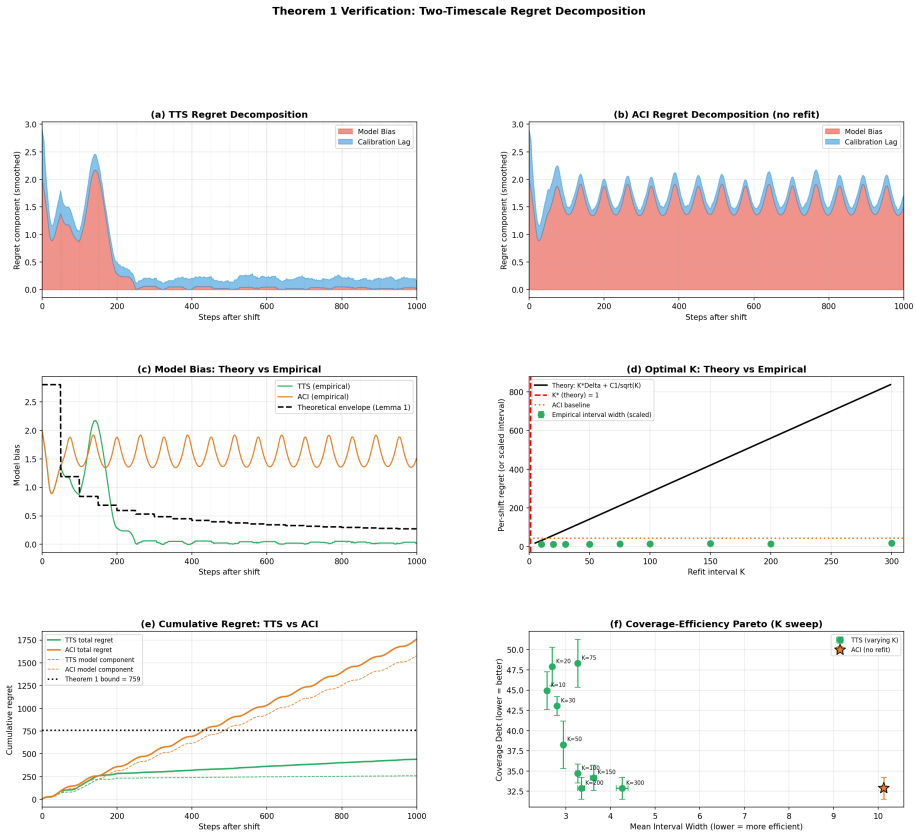

Figure 1: Theorem 1 verification. (a,b) Regret decomposition into model bias (red) and calibration lag (blue) for TTS and ACI. TTS model bias resets at each refit; ACI's never vanishes. (c) Theoretical envelope (dashed) captures the empirical model bias decay. (d) Optimal K from theory (red dashed) matches the empirical minimum. (e) Cumulative regret: TTS stays below the theoretical bound. (f) Coverage-efficiency Pareto front over K sweep.